1. Clinical Question & Core Hypothesis

Clinical Question

BCG elicits local anti-tumor immunity in the urothelium, yet some tumors still recur/metastasize — why do immune-recruiting treatments (BCG) fail in certain cases? The Ontologist highlights EMT-driven metastatic potential and Wnt/beta-catenin (CTNNB1) as a core axis, suggesting EMT plasticity might interfere with BCG-induced immunity. 2) Tier 1 paths used: (a) EMT → metastatic spread (provides the core problem: EMT enables dissemination), and (b) Wnt/beta-catenin → EMT (CTNNB1 as the molecular driver that can enforce EMT programs). I built on these because they form a clear mechanistic chain linking signaling to phenotype to clinical outcome. 3) Tier 2 supporting evidence: CDH1 loss and SNAI1/ZEB1-mediated partial-EMT states, plus MMP9-driven matrix remodeling, biologically plausibly modulate immune access/function and radiotherapy sensitivity; these reinforce the mechanism that EMT programs can both alter immune infiltration patterns and treatment susceptibility. 4) Endpoint choice rationale: I selected therapy_response (BCG response) because the ontologist emphasized BCG as the actionable clinical modality; testing how EMT/Wnt signaling modulates BCG efficacy is a focused, testable, and clinically actionable question (more direct than OS/PFI given BCG-specific mechanism). 5) Deliberate exclusions: I excluded Tier 3/irrelevant items (the breast-derived cohort edge and generic pathway edges) because they are not urothelial-specific; I placed Tier 2 genes (CDH1, SNAI1, ZEB1, MMP9) into secondary_modifiers rather than the core hypothesis to keep the claim sharply focused on CTNNB1-driven EMT and immune (TIL) interactions. I also excluded claims about radiotherapy and systemic immunotherapies to avoid conflating endpoints and mechanisms?

Working Hypothesis

Primary hypothesis: CTNNB1-driven activation of Wnt/beta-catenin signaling enforces EMT programs in urothelial tumors, creating a tumor microenvironment in which intravesical BCG-triggered anti-tumor immunity is impaired, leading to reduced therapy_response. Ancillary hypothesis: EMT status modulates the relationship between TIL density and BCG response such that tumors with high TIL density but concurrent high CTNNB1 activity and EMT signatures exhibit poorer BCG response than tumors with low CTNNB1 activity or low EMT signaling. Mechanistically, CTNNB1 activation reprograms chemokine/cytokine profiles and antigen presentation pathways (e.g., reduced T cell recruiting chemokines, altered MHC expression) and promotes T cell exclusion or dysfunction, diminishing local immune cascade initiation by BCG. Clinically, CTNNB1-driven EMT status plus high TIL density will stratify patients into distinct BCG response groups, independent of tumor mutational burden alone. Secondary modifiers (CDH1, SNAI1, ZEB1, MMP9) may modulate the magnitude of immune evasion but are not essential drivers of the primary CTNNB1–EMT–immune interaction; their inclusion will refine risk stratification and mechanistic nuance.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CTNNB1, EMT_status
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CTNNB1: CTNNB1) as a core axis, suggesting EMT plasticity might interfere with BCG-induced immunity.

EMT_status: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CDH1	Supporting modifier from knowledge graph
Secondary	SNAI1	Supporting modifier from knowledge graph
Secondary	ZEB1	Supporting modifier from knowledge graph
Secondary	MMP9	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , and innate sensing pathways after BCG-mimetic stimulation. - Co-culture assays: CTNNB1-active vs. control cells with human T cells to evaluate T cell activation, proliferation, and cytotoxicity in the presence of BCG-like stimuli. - In vivo: mouse urothelial cancer models with inducible CTNNB1 activity, treated with intravesical BCG, to measure tumor recurrence, TIL localization, and cytokine milieus; include spatial analyses to assess T cell exclusion patterns. - Human data analyses: retrospective/prospective analyses of urothelial cancer cohorts with available CTNNB1/EMT signatures and documented BCG outcomes; assess correlations between CTNNB1 activity, EMT scores, TIL density/location, and therapy_response. - Tissue-based analyses: IHC/IF for beta-catenin (nuclear localization), E-cadherin, vimentin, and T cell markers; correlate with BCG response data. - Advanced profiling: spatial transcriptomics or single-cell RNA-seq to map CTNNB1-driven EMT programs to immune cell spatial arrangements and activation states.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CDH1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SNAI1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.